EEES302 Digital Design
Experiment 7
Counters

Purpose:

The objective of this experiment is to explore the operational principles of digital counters and to gain hands-on experience by designing, and building a synchronous counter.

Review of Experiment:

Counters are essential sequential logic circuits used in computers and other digital systems to count events, track time, and manage state transitions.

• Binary counters: These circuits sequence through a standart binary progression. In an asynchronous counter, only the first flip-flop receives the external clock, while subsequent flip-flops are triggered by the output of the preceding one. In a synchronous counter, all flip-flops are driven simultaneously by a common clock.

• Decade counters: These circuits count from 0 to 9 and recyle back to 0. This is achieved by taking a standart 4-bit binary counter and employing logic gates to truncate the sequence.

• Up-Down Counters: These highly versatile synchronous counters include a dedicated control pin that dictates the counting direction.

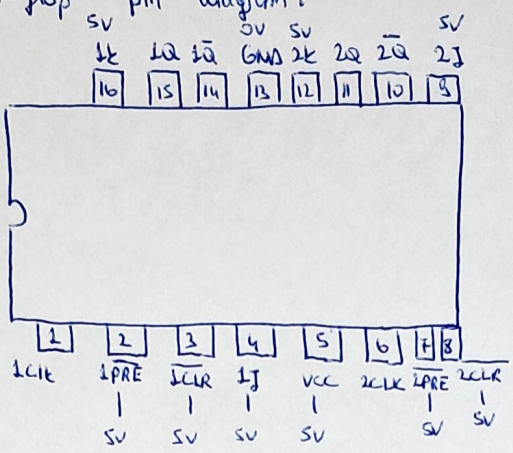
Equipment List:

- 74LS76 JK flip-flop
- Jumper wires
- Bread board

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7476 Jk flip-flop pin diagram:

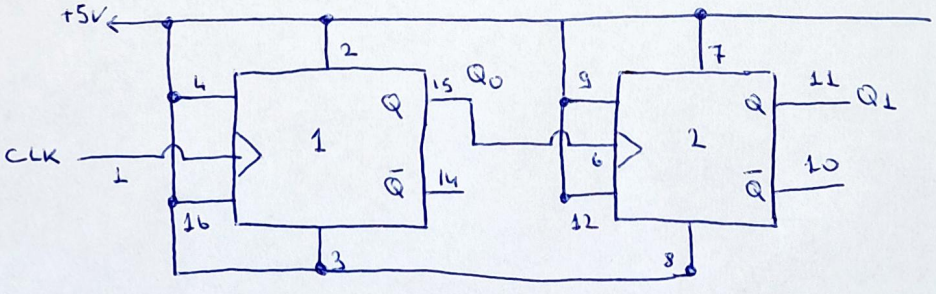


Experimental Procedure:

Decimal State	Present State	Next State	J_1	k_1	J_2	k_2
0	00	01	1	1	1	1
1	01	10	1	1	1	1
2	10	11	1	1	1	1
3	11	00	1	1	1	1

Jk Characteristic Table		
J	k	Q
0	0	Q → Hold
0	1	0 → Reset
1	0	1 → Set
1	1	$\bar{Q}$ → Toggle

Circuit Diagram:



Based on the circuit diagram it remains toggle state. The state changes are driven by the signal generators applied signal. We applied a 1Hz 5V square wave and observed the output Q_0 and Q_1.

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Conclusion:

The experiment was succesfully done. A custom-sequence synchronous counter was designed and verified using 7476 Dual Jk flip-flop circuit.

By mapping the desired 1-4 loop into binary states and solving for the Jk excitation table requirements, the counter reliably produced the intended sequence.

This experiment practically reinforced theoretical knowledge of sequential logic, state transition table, and the specific handling of edge triggered elements

Enrollment Year
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